Tutorial 1: Infinite Sequences and Series

1 September 2025

1. Find the limit of the following sequence:

$$\sqrt{2},\ \sqrt{2\sqrt{2}},\ \sqrt{2\sqrt{2\sqrt{2}}},\ \sqrt{2\sqrt{2\sqrt{2\sqrt{2}}}},\ \ldots.$$

Observe that we may write the terms of the sequence as

$$\begin{aligned}
a_1 &= 2^{1/2}, \\
a_2 &= (2 \cdot a_1)^{1/2} = \left(2^{3/2}\right)^{1/2} = 2^{3/4}, \\
a_3 &= (2 \cdot a_2)^{1/2} = \left(2^{7/4}\right)^{1/2} = 2^{7/8}, \\
a_4 &= (2 \cdot a_3)^{1/2} = \left(2^{15/8}\right)^{1/2} = 2^{15/16}, \\
&\vdots
\end{aligned}$$

In particular, the nth term is given by

$$a_n = (2 \cdot a_{n-1})^{1/2} = 2^{(2^n-1)/2^n} = 2^{1-(1/2)^n}.$$

Hence, we find that

$$\lim_{n\to\infty} 2^{1-(1/2)^n} = 2^{1-0} = 2.$$

2. Determine whether each of the following sequences $\{a_n\}$ converges or diverges. If the sequence converges, find its limit.

   (a) $a_n = \tan\left(\dfrac{2n\pi}{1+8n}\right)$

   (b) $a_n = \dfrac{n^5}{\sqrt{n^6+3n^2}}$

   (c) $a_n = \left(1+\dfrac{k}{n}\right)^n$

(a) We compute the limit

$$\lim_{n\to\infty} \tan\left(\frac{2n\pi}{1+8n}\right) = \tan\left(\lim_{n\to\infty} \frac{2n\pi}{1+8n}\cdot\frac{1/n}{1/n}\right) = \tan\left(\lim_{n\to\infty}\frac{2\pi}{1/n+8}\right)$$
$$= \tan\left(\frac{2\pi}{8}\right) = \tan\left(\frac{\pi}{4}\right) = 1.$$

Hence, the sequence converges to the limit 1.

(b) Dividing the expression throughout by n^5, we have

$$\lim_{n\to\infty}\frac{n^5}{\sqrt{n^6+3n^2}}\cdot\frac{1/n^5}{1/n^5} = \lim_{n\to\infty}\frac{1}{\sqrt{n^6/n^{10}+3n^2/n^{10}}}$$
$$= \lim_{n\to\infty}\frac{1}{\sqrt{1/n^4+3/n^8}} \to \infty.$$

Thus, the sequence diverges.

(c) Let

$$y = \left(1+\frac{k}{x}\right)^x \implies \ln y = x\ln\left(1+\frac{k}{x}\right).$$

We now compute the limit:

$$\lim_{x\to\infty}\ln y = \lim_{x\to\infty} x\ln\left(1+\frac{k}{x}\right) = \lim_{x\to\infty}\frac{\ln(1+k/x)}{1/x} \to \frac{0}{0}.$$

We can thus apply l'Hôpital's rule:

$$\lim_{x\to\infty}\frac{\ln(1+k/x)}{1/x} = \lim_{x\to\infty}\frac{\frac{1}{1+k/x}\cdot -\frac{k}{x^2}}{-1/x^2} = \lim_{x\to\infty}\frac{k}{1+k/x} = k.$$

Hence, we can conclude that

$$\lim_{x\to\infty}\left(1+\frac{k}{x}\right)^x = \lim_{x\to\infty} e^{\ln y} = e^k.$$

The corresponding sequence thus converges.

3. The nth partial sum of a series is given by

$$\sum_{i=1}^{n} a_i = \frac{n-1}{n+1}.$$

(a) Find an expression for the nth term of the sequence a_n.

(b) Hence, compute the sum of the infinite series.

We list out the first few partial sums:

$$\begin{aligned} a_1 &= \frac{1-1}{1+1} = 0 \\ a_1 + a_2 &= \frac{2-1}{2+1} = \frac{1}{3} \\ a_1 + a_2 + a_3 &= \frac{3-1}{3+1} = \frac{1}{2} \\ a_1 + a_2 + a_3 + a_4 &= \frac{4-1}{4+1} = \frac{3}{5} \\ &\vdots \end{aligned}$$

Thus, we have

$$a_1 = 0, \quad a_2 = \frac{1}{3}, \quad a_3 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}, \quad a_4 = \frac{3}{5} - \frac{1}{2} = \frac{1}{10}, \quad \ldots.$$

More generally,

$$\begin{aligned} a_n &= \sum_{i=1}^{n} a_i - \sum_{i=1}^{n-1} a_i = \frac{n-1}{n+1} - \left(\frac{(n-1)-1}{(n-1)+1}\right) \\ &= \frac{n-1}{n+1} - \frac{n-2}{n} = \frac{\left(n^2-n\right) - \left(n^2-n-2\right)}{n\,(n+1)} = \frac{2}{n\,(n+1)}. \end{aligned}$$

The sum of the infinite series is simply

$$\lim_{n\to\infty} \sum_{i=1}^{n} a_i = \lim_{n\to\infty} \frac{n-1}{n+1} \cdot \frac{1/n}{1/n} = \lim_{n\to\infty} \frac{1-1/n}{1+1/n} = 1.$$

4. Let $x = 0.\bar{9} = 0.99999\ldots$.

(a) Express x as a sum of an infinite geometric series.

(b) Hence, or otherwise, express x as a rational number.

Observe that

$$\begin{aligned} x &= 0.9 + 0.09 + 0.009 + 0.0009 + 0.00009 + \cdots \\ &= 0.9 + 0.9 \cdot 0.1 + 0.9 \cdot 0.1^2 + 0.9 \cdot 0.1^3 + 0.9 \cdot 0.1^4 + \cdots. \end{aligned}$$

In particular, we have the infinite sum

$$x = \sum_{n=0}^{\infty} 0.9 \cdot (0.1)^n.$$

This is a geometric series with first term $a = 0.9$ and common ratio $r = 0.1$. Hence,

$$x = \frac{0.9}{1-0.1} = \frac{0.9}{0.9} = 1.$$

5. Explain why each of the following geometric series converges, and compute their sum.

   (a) $2 + 0.5 + 0.125 + 0.03125 + \cdots$

   (b) $\displaystyle\sum_{n=0}^{\infty} \frac{e^n}{5^{n+1}}$

   (c) $\displaystyle\sum_{n=1}^{\infty} \frac{(-4)^{n-1}}{6^n}$

(a) Observe that the sum is of the form

$$2 + 2 \cdot \frac{1}{4} + 2 \cdot \left(\frac{1}{4}\right)^2 + 2 \cdot \left(\frac{1}{4}\right)^3 + \cdots.$$

In particular, we have a geometric series with first term $a = 2$ and common ration $r = 1/4$. Since $|r| < 1$, the geometric series must converge to the sum

$$\frac{a}{1-r} = \frac{2}{1-1/4} = \frac{8}{3}.$$

(b) We may rewrite the sum as

$$\sum_{n=0}^{\infty} \frac{e^n}{5^{n+1}} = \sum_{n=0}^{\infty} \frac{1}{5} \cdot \left(\frac{e}{5}\right)^n,$$

which is a geometric series with first term $a = 1/5$ and common ratio $r = e/5 \approx 0.54366$. Since $|r| < 1$, the geometric series must converge to the sum

$$\frac{a}{1-r} = \frac{1/5}{1-e/5} = \frac{1}{5-e}.$$

(c) We may rewrite the sum as

$$\sum_{n=1}^{\infty} \frac{(-4)^{n-1}}{6^n} = \sum_{n=1}^{\infty} \left(-\frac{1}{4}\right)\left(-\frac{2}{3}\right)^n.$$

This is a geometric series with first term and common ratio respectively given by

$$a = -\frac{1}{4} \cdot -\frac{2}{3} = \frac{1}{6}, \quad r = -\frac{2}{3}.$$

Since $|r| < 1$, the geometric series must converge to the sum

$$\frac{a}{1-r} = \frac{1/6}{1-(-2/3)} = \frac{1}{10}.$$

6. Determine the values of x for which each of the following geometric series converges. For those values of x, compute the corresponding sum of the series.

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{(x-2)^n}{3^{n+1}}$

(b) $\displaystyle\sum_{n=0}^{\infty} e^{nx}$

(c) $\displaystyle\sum_{n=0}^{\infty} \frac{\sin^n x}{3^n}$

(a) We may rewrite the sum as

$$\sum_{n=1}^{\infty} \frac{(x-2)^n}{3^{n+1}} = \sum_{n=1}^{\infty} \frac{1}{3} \cdot \left(\frac{x-2}{3}\right)^n.$$

The first term and common ratio are given by

$$a = \frac{1}{3} \cdot \frac{x-2}{3} = \frac{x-2}{9}, \quad r = \frac{x-2}{3}.$$

Thus, the infinite series converges when

$$-1 < \frac{x-2}{3} < 1 \implies -3 < x-2 < 3 \implies -1 < x < 5.$$

For $-1 < x < 5$, the sum of the infinite series is

$$\frac{a}{1-r} = \frac{(x-2)/9}{1-(x-2)/3} = \frac{(x-2)/9}{(5-x)/3} = \frac{1}{3} \cdot \frac{x-2}{5-x}.$$

(b) We may rewrite the sum as

$$\sum_{n=0}^{\infty} e^{nx} = \sum_{n=0}^{\infty} (e^x)^n.$$

The first term and common ratio are given by $a = (e^x)^0 = 1$ and $r = e^x$. Thus, the infinite series converges when

$$-1 < e^x < 1.$$

Since e^x is always positive, we simply require that $x < 0$. In this case, the sum of the infinite series is

$$\frac{a}{1-r} = \frac{1}{1-e^x}.$$

(c) We may rewrite the sum as

$$\sum_{n=0}^{\infty} \frac{\sin^n x}{3^n} = \sum_{n=0}^{\infty} \left(\frac{\sin x}{3}\right)^n.$$

The first term and common ratio are given by

$$a = \left(\frac{\sin x}{3}\right)^0 = 1, \quad r = \frac{\sin x}{3},$$

and the infinite series converges when

$$-1 < \frac{\sin x}{3} < 1 \quad \implies \quad -3 < \sin x < 3.$$

Note that the inequality holds for all values of x; hence, the series always converges to the sum

$$\frac{a}{1-r} = \frac{1}{1-(\sin x)/3} = \frac{3}{3-\sin x}.$$

7. Find a formula for the nth partial sum of each of the following series. Hence, determine if the series converges or diverges. If a series converges, find its sum.

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$

(b) $\displaystyle\sum_{n=1}^{\infty} \ln\left(\frac{n+1}{n}\right)$

(c) $\displaystyle\sum_{n=1}^{\infty} \left(\cos\frac{1}{n^2} - \cos\frac{1}{(n+1)^2}\right)$

(a) Observe that the terms of the sum may be written as

$$\frac{1}{n(n+1)} = \frac{1+n-n}{n(n+1)} = \frac{n+1}{n(n+1)} - \frac{n}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}.$$

Thus, the nth partial sum is given by

$$\sum_{i=1}^{n} \frac{1}{i} - \frac{1}{i+1} = \left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right) = 1 - \frac{1}{n+1}.$$

Observe that

$$\lim_{n\to\infty}\left(\sum_{i=1}^{n} \frac{1}{n(n+1)}\right) = \lim_{n\to\infty} 1 - \frac{1}{n+1} = 1,$$

and the series converges.

(b) The terms of the sum may be written as

$$\ln\left(\frac{n+1}{n}\right) = \ln(n+1) - \ln n.$$

Thus, the nth partial sum is given by

$$\begin{aligned}\sum_{i=1}^{n} \ln(i+1) - \ln i &= (\ln 2 - \ln 1) + (\ln 3 - \ln 2) + (\ln 4 - \ln 3) + \cdots \\ &= -\ln 1 + \ln(n+1) = \ln(n+1).\end{aligned}$$

We compute the limit of the partial sums:

$$\lim_{n\to\infty}\left(\sum_{i=1}^{n} \ln\left(\frac{i+1}{i}\right)\right) = \lim_{n\to\infty} \ln(n+1) \to \infty.$$

Thus, the series diverges.

(c) The nth partial sum is given by

$$\begin{aligned}\sum_{i=1}^{n}\left(\cos\frac{1}{i^2} - \cos\frac{1}{(i+1)^2}\right) &= \left(\cos\frac{1}{1} - \cos\frac{1}{4}\right) + \left(\cos\frac{1}{4} - \cos\frac{1}{9}\right) + \cdots \\ &= \cos 1 - \cos\frac{1}{(n+1)^2}.\end{aligned}$$

We compute the limit of the partial sums:

$$\begin{aligned}\lim_{n\to\infty}\left(\sum_{i=1}^{n}\left(\cos\frac{1}{i^2} - \cos\frac{1}{(i+1)^2}\right)\right) &= \lim_{n\to\infty}\left(\cos 1 - \cos\frac{1}{(n+1)^2}\right) \\ &= \cos 1 - \cos 0 = \cos 1 - 1.\end{aligned}$$

Thus, the series converges.

8. Determine the Maclaurin series of each of the following functions.

   (a) $f(x) = x^2/\sqrt{2+x}$

   (b) $f(x) = x^2 \ln\left(1+x^3\right)$

   (c) $f(x) = \sin^2 x$

   Hint: You may use any of the results on Maclaurin series we have already derived in class.

   (a) We can use the binomial series to determine the expansion of $f(x)$. First, note that

$$(2+x)^{-1/2} = \left(2\left(1+\frac{x}{2}\right)\right)^{-1/2} = \frac{1}{\sqrt{2}}\left(1+\frac{x}{2}\right)^{-1/2}.$$

   Thus,

$$f(x) = \frac{x^2}{\sqrt{2}}\left(1+\frac{x}{2}\right)^{-1/2} = \frac{x^2}{\sqrt{2}}\sum_{n=0}^{\infty}\binom{-1/2}{n}\left(\frac{x}{2}\right)^n.$$

   We can write out the first few terms of this series:

$$\begin{aligned}
f(x) &= \frac{x^2}{\sqrt{2}}\left(1+\left(-\frac{1}{2}\right)\left(\frac{x}{2}\right)+\frac{-\frac{1}{2}\cdot-\frac{3}{2}}{2!}\left(\frac{x}{2}\right)^2+\frac{-\frac{1}{2}\cdot-\frac{3}{2}\cdot-\frac{5}{2}}{3!}\left(\frac{x}{2}\right)^3+\cdots\right)\\
&= \frac{x^2}{\sqrt{2}}+\frac{x^2}{\sqrt{2}}\left(\frac{-1}{2^1\cdot 2^1}x+\frac{-1\cdot-3}{2^2\cdot 2!\cdot 2^2}x^2+\frac{-1\cdot-3\cdot-5}{2^3\cdot 3!\cdot 2^3}x^3+\cdots\right)\\
&= \frac{x^2}{\sqrt{2}}+\frac{x^2}{\sqrt{2}}\left(\sum_{n=1}^{\infty}\frac{(-1)^n\left(1\cdot 3\cdot 5\cdot\cdots\cdot(2n-1)\right)}{n!2^{2n}}x^n\right)\\
&= \frac{x^2}{\sqrt{2}}+\sum_{n=1}^{\infty}\frac{(-1)^n\left(1\cdot 3\cdot 5\cdot\cdots\cdot(2n-1)\right)}{n!2^{2n+1/2}}x^{n+2}.
\end{aligned}$$

   (b) We use the Maclaurin series of $\ln(1+x)$: observe that

$$\ln\left(1+x^3\right) = \sum_{n=1}^{\infty}(-1)^{n-1}\frac{\left(x^3\right)^n}{n} = \left(x^3\right)-\frac{\left(x^3\right)^2}{2}+\frac{\left(x^3\right)^3}{3}-\frac{\left(x^3\right)^4}{4}+\cdots$$

   Hence,

$$f(x) = x^2\ln\left(1+x^3\right) = x^2\sum_{n=1}^{\infty}(-1)^{n-1}\frac{x^{3n}}{n} = \sum_{n=1}^{\infty}(-1)^{n-1}\frac{x^{3n+2}}{n}.$$

   (c) We may rewrite the function as

$$f(x) = \sin^2 x = \frac{1}{2}(1-\cos 2x).$$

   We can thus use the Maclaurin series of cosine to determine the expansion of $f(x)$. Note that

$$\cos 2x = \sum_{n=0}^{\infty}(-1)^n\frac{(2x)^{2n}}{(2n)!} = 1-\frac{(2x)^2}{2!}+\frac{(2x)^4}{4!}-\frac{(2x)^6}{6!}+\cdots$$

   Hence,

$$\begin{aligned}
\frac{1}{2}(1-\cos 2x) &= \frac{1}{2}\left(1-\left(1-\frac{(2x)^2}{2!}+\frac{(2x)^4}{4!}-\frac{(2x)^6}{6!}+\cdots\right)\right)\\
&= \frac{1}{2}\left(\frac{2^2x^2}{2!}-\frac{2^4x^4}{4!}+\frac{2^6x^6}{6!}+\cdots\right)\\
&= \frac{1}{2}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}2^{2n}x^{2n}}{(2n)!} = \sum_{n=1}^{\infty}\frac{(-1)^{n+1}2^{2n-1}x^{2n}}{(2n)!}.
\end{aligned}$$

9. Use an appropriate Maclaurin series to evaluate the following sums.

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{2^n n!}$

(b) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n}{2^n n}$

(c) $\displaystyle 1 - \ln 2 + \frac{(\ln 2)^2}{2!} - \frac{(\ln 2)^3}{3!} + \cdots$

(a) We may rewrite the sum as

$$\sum_{n=1}^{\infty} \frac{1}{2^n n!} = \sum_{n=1}^{\infty} \frac{(1/2)^n}{n!}.$$

This is simply the Maclaurin series expansion of e^x, with $x = 1/2$ and with the first term omitted:

$$\sum_{n=1}^{\infty} \frac{1}{2^n n!} = \sum_{n=1}^{\infty} \frac{(1/2)^n}{n!} = \left(\sum_{n=0}^{\infty} \frac{(1/2)^n}{n!}\right) - 1 = \sqrt{e} - 1.$$

(b) We may rewrite the sum as

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{2^n n} = \sum_{n=1}^{\infty} (-1)^n \frac{(1/2)^n}{n} = -\sum_{n=1}^{\infty} (-1)^{n-1} \frac{(1/2)^n}{n}.$$

We may use the Maclaurin series expansion of $\ln(1+x)$, with $x = 1/2$:

$$-\sum_{n=1}^{\infty} (-1)^{n-1} \frac{(1/2)^n}{n} = -\ln\left(1 + \frac{1}{2}\right) = -\ln\frac{3}{2}.$$

(c) Observe that we may express the sum as

$$1 - \ln 2 + \frac{(\ln 2)^2}{2!} - \frac{(\ln 2)^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{(-\ln 2)^n}{n!}.$$

We may thus directly use the Maclaurin series of e^x, with $x = -\ln 2$:

$$\sum_{n=0}^{\infty} \frac{(-\ln 2)^n}{n!} = e^{-\ln 2} = \frac{1}{2}.$$

10. Let
$$f(x) = \frac{1}{\sqrt{1-x^2}}.$$

(a) Use the binomial series to express $f(x)$ as an infinite series.

(b) Hence, or otherwise, find the Maclaurin series for $\sin^{-1} x$.

(a) Observe that
$$f(x) = \left(1-x^2\right)^{-1/2}.$$
Thus, by the binomial series, we have
$$\left(1-x^2\right)^{-1/2} = \sum_{n=0}^{\infty} \binom{-1/2}{n} \left(-x^2\right)^n = \sum_{n=0}^{\infty} \binom{-1/2}{n} (-1)^n \left(x^{2n}\right).$$

(b) Let us compute the derivative of the inverse sine function: let
$$y = \sin^{-1} x \quad \Longrightarrow \quad \sin y = x.$$
Differentiating both sides yields
$$\cos y \cdot \frac{dy}{dx} = 1 \quad \Longrightarrow \quad \frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-\sin^2 y}} = \frac{1}{\sqrt{1-x^2}}.$$
Hence,
$$\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}.$$
We can thus obtain the Maclaurin series for $\sin^{-1} x$ by integrating the Maclaurin series of $f(x)$:
$$\begin{aligned}
\sin^{-1} x &= \int \sum_{n=0}^{\infty} \binom{-1/2}{n} (-1)^n \left(x^{2n}\right) \, dx \\
&= \sum_{n=0}^{\infty} \binom{-1/2}{n} (-1)^n \int \left(x^{2n}\right) \, dx \\
&= \left[\sum_{n=0}^{\infty} \binom{-1/2}{n} (-1)^n \left(\frac{x^{2n+1}}{2n+1} \right) \right] + c.
\end{aligned}$$
Note that when $x = 0$, we have
$$\sin^{-1}(0) = \sum_{n=0}^{\infty} \binom{-1/2}{n} (-1)^n \left(\frac{0^{2n+1}}{2n+1} \right) + c \quad \Longrightarrow \quad c = 0.$$
Hence, we have the Maclaurin series expansion
$$\sin^{-1} x = \sum_{n=0}^{\infty} \binom{-1/2}{n} (-1)^n \left(\frac{x^{2n+1}}{2n+1} \right) = x + \frac{x^3}{6} + \frac{3x^5}{40} + \frac{5x^7}{112} + \cdots.$$